

Project Proposal

POCKETCAM

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PocketCam: Design and Development of a Portable ESP32-Based Digital Camera with Offline Wireless Photo Sharing

Project Overview

PocketCam is a portable digital camera built on the ESP32 platform. The core idea is a handheld device that captures real photographs and saves them to a microSD card using a physical shutter button, similar to a simple point and shoot camera. On top of that core function, the project adds embedded and networking features that make it useful in a real social setting, specifically an offline Wi-Fi gallery that lets friends view and download photos directly from the camera without any internet connection or cloud service.

The project is designed to be a genuine personal use case first and a portfolio project second. It should be something Stel actually wants to carry and use when going out with friends, while also demonstrating strong technical depth across embedded systems, networking, firmware design and hardware integration

Motivation

The idea comes from wanting a dedicated device for capturing memories with friends that is more intentional than a phone camera, while also creating an engineering project with enough real constraints to be technically demanding. Working within the memory, processing and storage limits of a small microcontroller, while still shipping a usable product, is where most of the learning happens.

MVP Scope

The following features form the minimum viable product. Everything else is a stretch goal and will only be attempted once these are reliable.

- Physical shutter button that reliably captures a photo on press
- Photos saved as numbered JPEG files to a microSD card
- Rechargeable battery operation with a power switch
- Onboard status indicator showing ready, capturing and saved states
- Local Wi-Fi access point hosted by the camera itself
- A simple web gallery page that lists photos and allows downloading them from a connected phone

Stretch Goals (After MVP Is Working)

These are kept clearly separate from the MVP so scope does not creep before the core is solid. They are ordered roughly by how much value they add relative to effort.

- Photo booth mode with a short countdown sequence for multiple shots
- Surprise mode that locks the gallery until a chosen time or event
- Session based photo albums, for example one folder per outing
- Simple colour filters such as warm, cool, monochrome and vintage
- QR code on the enclosure for quick Wi-Fi connection
- Battery level monitoring shown on the status display
- Burst mode with a short sequence of photos on a long press
- Photo metadata log including date, time, mode and filter used

At most two or three of these should be attempted for version one. The rest can be documented as future work in the final write up.

Hardware Component List

Components	Purpose	Notes
ESP32-CAM (AI-Thinker)	Main controller and camera module	Confirm GPIO availability before wiring extra buttons or sensors
MicroSD card and slot	Photo storage	Many ESP32-CAM boards include a built in slot
Physical shutter button	Trigger photo capture	Simple momentary push button
LiPo battery and TP4056 module	Portable rechargeable power	Confirm charging and discharge current limits
Status LED or small display	Show ready, capturing and saved states	OLED optional if GPIO allows it
Power switch	Turn device on and off	Should not require removing the battery
Custom enclosure	Handheld housing	Design after core electronics are validated

Milestone Plan

Milestone	Goal	Exit Criteria
1. Hardware confirmation	Finalise board, storage, power and wiring plan	Full component list confirmed and ordered
2. Core capture loop	Button press saves a numbered JPEG to SD card	Repeated captures work without crashing or corrupting files
3. Power and enclosure basics	Batter, switch and status indicator working together	Device runs unthethered for a full test session
4. Wi-Fi gallery	Camera hosts its own access point and gallery webpage	A phone can connect, view and download photos with no internet
5. One Stretch feature	Add a single polished extra feature	Feature works reliably in a live demo, not just in testing
6. Enclosure and polish	Finish physical housing and final testing	Device is demo ready and safe to hand yo someone else to use

Risks and Constrains

- The ESP32-CAM has limited RAM, so running the camera and a web server at the same time needs careful memory management

- SD card write speed may limit how fast photos can be taken in sequence, which affects any burst style feature
- Most GPIO pins are already used by the camera module, so extra buttons, LEDs or sensors need to be planned around what remains free
- Without internet access, accurate timestamps need either a real time clock module or a sync step when a phone connects
- Enclosure design and printing takes real time and should not be left until the very end

Documentation Plan

A build log will be kept from the first hardware test onward, recording problems encountered and how they were solved. This will live in the project GitHub repository alongside the code, and will form the basis of the final case study summary used in internship and job applications.

Target Outcome Statement

By the end of this project, the goal is to be able to describe it as follows: designed and developed a portable embedded digital camera using the ESP32 platform, supporting local JPEG storage, physical camera controls, rechargeable operation and an offline Wi-Fi gallery that allows multiple users to access and download images without any cloud service.